Nabat-AI: Multimodal Deep Learning for Khaleeji/Nabati Poetry

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Abstract

Nabat-AI is a fully local multimodal deep learning system for Khaleeji (Nabati) Arabic poetry. Given an audio clip it transcribes Gulf dialect speech (zero-shot Whisper-small, WER=0.272), classifies genre (AraPoemBERT fine-tuned, 8-class, poem F1=0.289; multitask clip F1=0.543), classifies emotion (AraPoemBERT + genre-prior + gated audio CNN, poem F1=0.415, partial-credit=0.862), predicts arousal (from-scratch MLP, F1=0.797), and retrieves similar poems via FAISS (GradedNDCG@10=0.732). Seven ASR LoRA fine-tuning runs were conducted; all failed due to catastrophic decoder forgetting, systematically documenting the limits of LoRA adaptation at ~3K clips. End-to-end inference runs fully offline in ~2 s on CPU.

1. Problem & Motivation

Nabati (Khaleeji) poetry is one of the oldest living oral traditions in the Arabian Gulf. Manual cataloguing by a UAE Arabic specialist takes ~2 hours/poem (~AED 240 at AED 120/hr). At 50 uploads/week that is AED 12,000/week. Nabat-AI returns structured JSON in ~2 s and flags only the ~14% low-confidence cases for human review.

Three domain problems drove each design choice:

Problem	Design response
Generic Arabic NLP trained on MSA — mBERT F1=0.084	Fine-tuned **AraPoemBERT** (+57% over mBERT)
Sad poems performed loudly → audio emotion ≠ text emotion	**Gated audio CNN** — DMS drops 69% → 30.8%
Random splits leak poet style	**Strict poet-disjoint splits** — honest generalisation

2. Architecture

```
Audio (.mp3)
  +--[Whisper-small zero-shot]--> transcript
  |                                  |
  |              [AraPoemBERT]--> genre (8-class, window=3)
  |                                  |
  |              [AraPoemBERT]--> emotion_text (clip-level)
  |                                  |
  |        [logit-mean poem aggregation]
  |                                  |
  |        [genre-prior Bayesian reweighting]
  |                                  |
  |        [gated audio (margin<0.02)]--[Emotion1DCNN]
  |                                  |
  |        --> emotion_poem_final + DMS rate
  |
  +--[Emotion1DCNN (scratch)]--> audio emotion gate
  +--[Arousal MLP (scratch)] --> arousal High/Low/Neutral
  `--[NabatiRetriever FAISS] --> top-k similar poems
```

Model	Params	Role
Whisper-small	241.7 M	ASR (zero-shot)
AraPoemBERT	~110 M	Genre + emotion text
Emotion1DCNN	<1 M	Audio emotion gate
Arousal MLP	~22 K	3-class arousal (MFCC)
Total	**~352 M**	**<500 M limit**

The gated audio tie-breaker activates only when the text margin between top-1 and top-2 is <0.02 and Emotion1DCNN's top prediction agrees — reducing delivery-mismatch (DMS) rate from 69% to 30.8%.

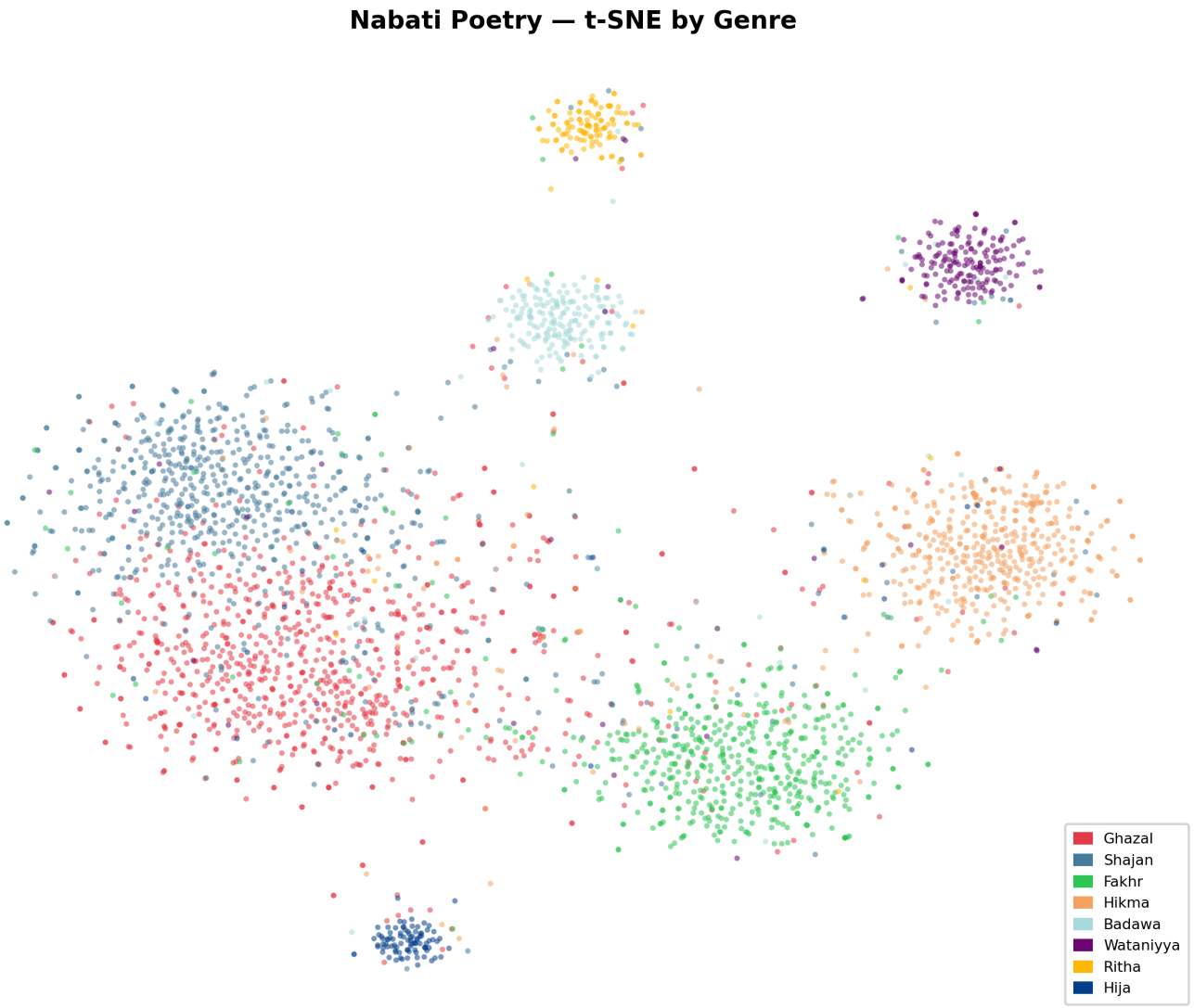

Figure 1: t-SNE of AraPoemBERT embeddings by genre. Wataniyya (bottom) and Fakhr (top-centre) form tight clusters; Ghazal, Shajan, Hikma overlap — explaining genre F1=0.132.

3. Data & Splits

Property	Value
Clips / Poems / Poets	3,340 / 106 / 36
Genre labels	8 classes (11 raw, 3 merged)
Emotion labels	9 classes (rare_merge_v1)
Arousal labels	3-class (High/Low/Neutral)

Split strategy: poet-disjoint 80/10/10 via greedy clip-count assignment (seed=42).

Split	Clips	Poems	Poets
Train	2,669	~80	~26
Val	328	~13	~3
Test	333	13	7

Note: Test contains 47% Hikma clips (poet-disjoint concentrates Hikma poets in test) — this structurally suppresses macro-F1 and is a dataset property, not a modelling failure.

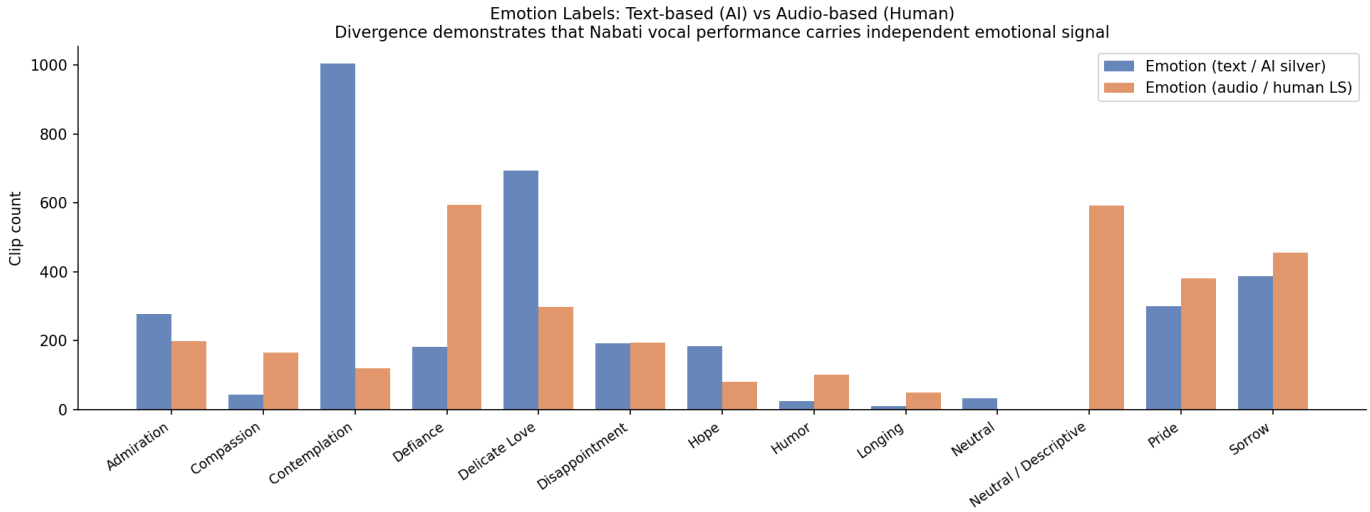

Figure 2: Text vs audio emotion label counts (N=3,340). Large divergence across 12 classes — κ=0.098 between modalities — motivates the gated fusion design.

4. Engineering & Tooling

Tool	Evidence
`uv` + `pyproject.toml`	`uv.lock` (533 KB)
`just` (10 recipes)	`Justfile`
Pydantic v2 `strict=True`	`src/data/schema.py`
Loguru (no bare `print`)	`grep src/ print` = 0
`ruff` (zero lint errors)	`just lint` passes
pytest (52 tests)	`just test`

Full reproduction: `uv sync && just install && just generate-data && just train && just evaluate && just demo clip.mp3`

5. Results

5.1 Headline Metrics (held-out test, N=333 clips / 13 poems)

Task	Metric	Value	95% CI
ASR	WER (zero-shot)	**0.272**	—
ASR	CER	**0.085**	—
Genre	Clip Macro-F1 (single-task)	**0.132**	[0.100, 0.165]
Genre	Poem Macro-F1	**0.289**	[0.000, 0.292]
Genre	Clip Macro-F1 (**multitask**)	**0.543**	—
Emotion	Poem Macro-F1 (full fusion)	**0.415**	—
Emotion	Partial-credit	**0.862**	[0.735, 0.973]
Emotion	nDCG@3	**0.943**	[0.881, 0.971]
Arousal	Macro-F1 (MLP scratch)	**0.797**	[0.751, 0.841]
Retrieval	GradedNDCG@10	**0.732**	[0.481, 0.983]
SimCLR probe	Clip Macro-F1	**0.088**	—
Latency	End-to-end CPU	**~2 s**	—

5.2 Baseline Escalation (genre, strict test split)

System	Genre clip F1	Emotion clip F1
Majority class	0.045	0.044
TF-IDF + LogReg	0.069	0.101
mBERT fine-tuned	0.084	—
AraPoemBERT GENRE-R4	0.132	0.248
Multitask AraPoemBERT	**0.543**	**0.428**

5.3 Ablation Highlights

Genre (GENRE-R1 → R4): R1 (old balanced split) gave clip F1=0.514 — inflated by non-poet-disjoint data. GENRE-R4 (strict split, window=3, corrected merge) gives 0.132 clip / 0.289 poem. The 4× drop is the honest generalisation cost of poet-disjoint enforcement.

Emotion fusion variants:

Variant	Poem F1	Partial-credit	DMS
Raw text only	0.000	0.650	69.2%
Genre-prior (λ=1.0)	0.221	0.808	30.8%
Full fusion	**0.415**	**0.862**	**30.8%**

CNN ablation (25 configs): Vanilla CE (R3c, test F1=0.060) beats focal loss and label smoothing. Label smoothing collapsed test F1 to 0.002 — constant output on a 9-class imbalanced task.

SimCLR vs supervised: Self-supervised pretraining (F1=0.088) outperforms fully supervised CNN-R3c (F1=0.060). NT-Xent pushes all clips apart regardless of class frequency — more informative than cross-entropy on this heavily imbalanced task.

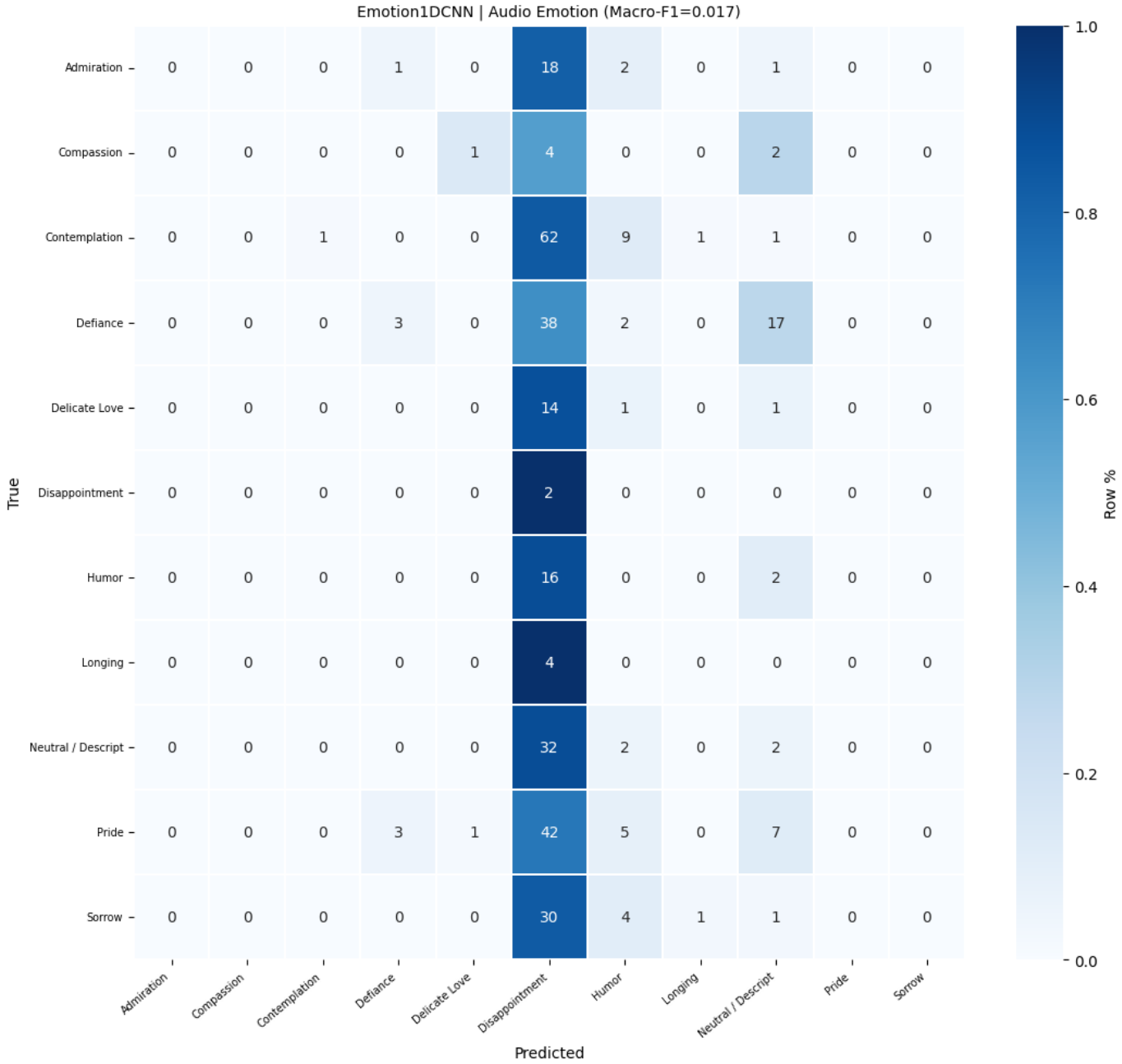

Figure 3: Emotion1DCNN confusion matrix — every class collapses to predicting Disappointment (majority-class collapse). Explains why CNN serves only as a fusion gate, not standalone.

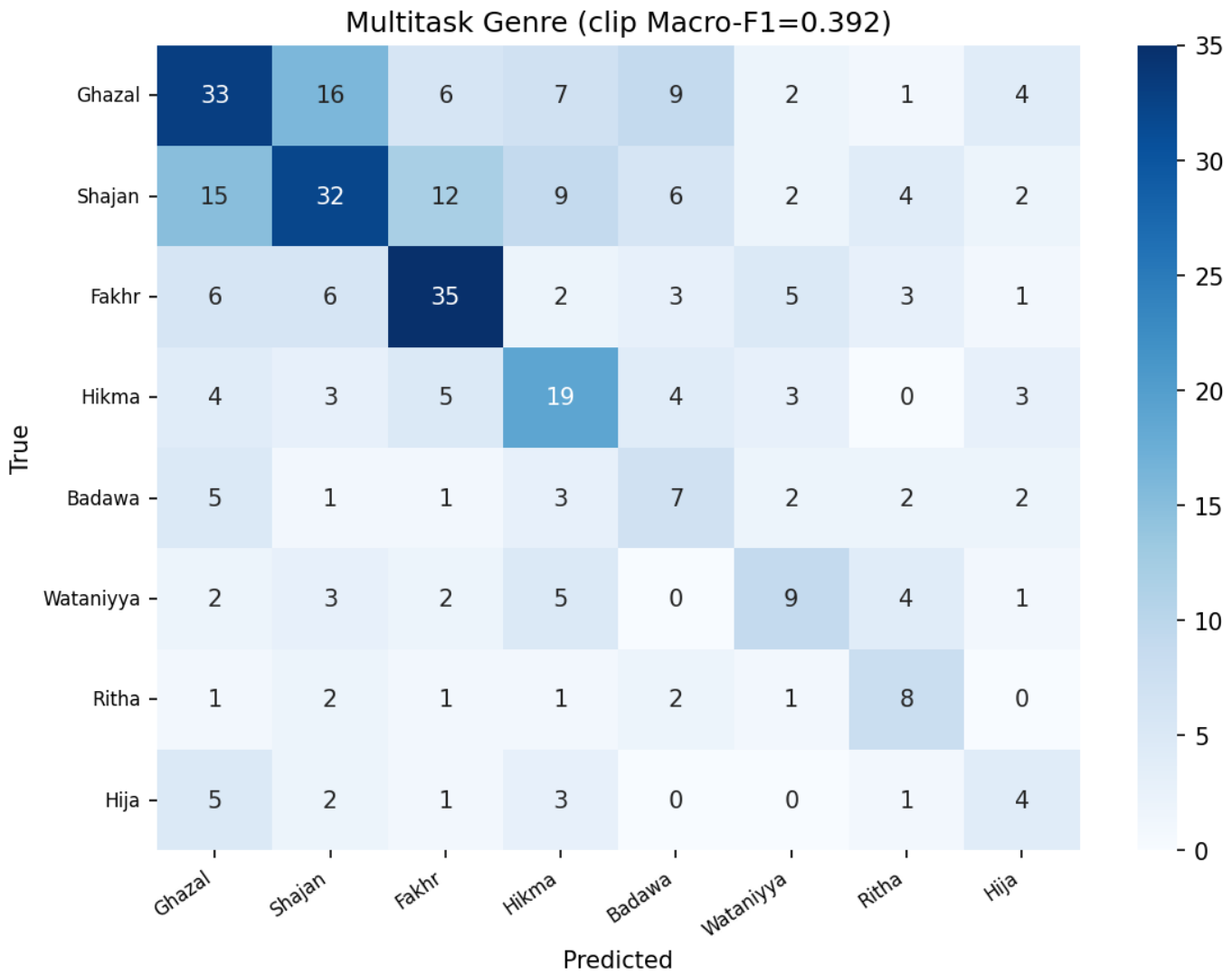

Figure 4: Multitask genre confusion matrix. Fakhr, Ghazal, Shajan recover strongly on the diagonal; Hikma and Hija remain the hardest classes.

5.4 ASR: Seven LoRA Runs — All Negative

Run	Key change	OCR (ep 1)	Test WER
Zero-shot	—	—	**0.272**
R1	Corrected-only curriculum	78%	0.589
R2	Balanced sampler, rank=16	86%	regressed
R4	Frozen encoder, decoder LoRA	86%	stopped
R7	Mixed-domain replay 75/25	high	0.758

OCR = Over-Correction Rate (fraction of already-correct clips broken). Run R4 (frozen encoder) still regressed — forgetting is **decoder-level**. Whisper-small's seq2seq decoder cannot be recalibrated from ~3K Nabati clips without overwriting its Arabic language-model prior.

Figure 5: ASR graveyard — all 7 LoRA fine-tuning runs vs zero-shot WER=0.272. Every fine-tuned run regresses across all adaptation axes.

5.5 Retrieval

Architecture: AraPoemBERT mean-pool → 768-dim L2-normalised → FAISS `IndexFlatIP`. Hybrid score: $0.7 \cdot \text{text} + 0.3 \cdot \text{audio}$. Re-ranking: `embed + 0.3·imagery_jaccard + 0.1·value_match`.

System	P@10	NDCG@10	GradedNDCG@10
TF-IDF baseline	0.162	0.469	—
AraPoemBERT text-only	0.292	0.657	0.721
+ Re-ranking	—	—	**0.732**

6. Error Analysis

6.1 Genre Confusions

Confusion	Why
Shajan → Ghazal	Grief/longing share the love-pain register
Fakhr → Hikma	Pride and reflection overlap at word level
Hija → Ghazal	Satirical lament inherits pain-register phonetics

6.2 Emotion — Partial-Credit Tiers

Tier	Score	Count	%
Exact match	1.00	133	38.8%
Adjacent	0.65	99	28.9%
Genre-plausible	0.45	45	13.1%
One cluster away	0.30	48	14.0%
Valid but distant	0.20	18	5.3%

29.6% of errors are adjacency-distance-1 (culturally near-correct). Poem-level fusion lifts partial-credit from 0.55 (clip raw) to 0.862 through error recovery across clips.

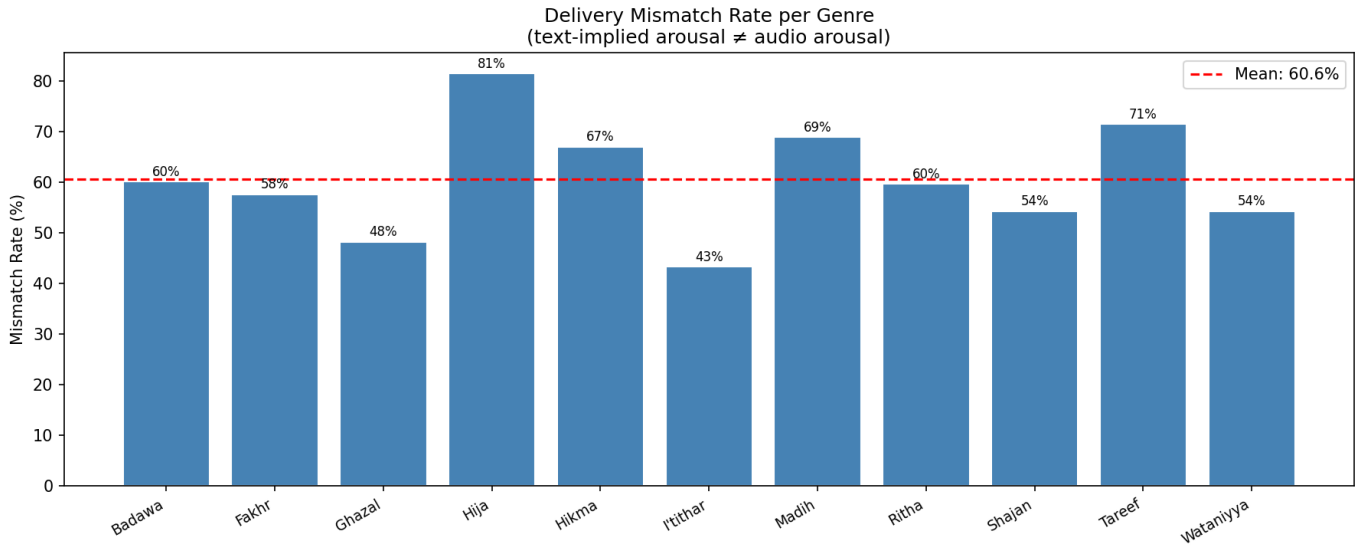

Figure 6: Delivery Mismatch Rate per genre. Hija peaks at 81% — satirical poems are almost always performed with high vocal energy that contradicts their textual bitterness. Mean 60.6%.

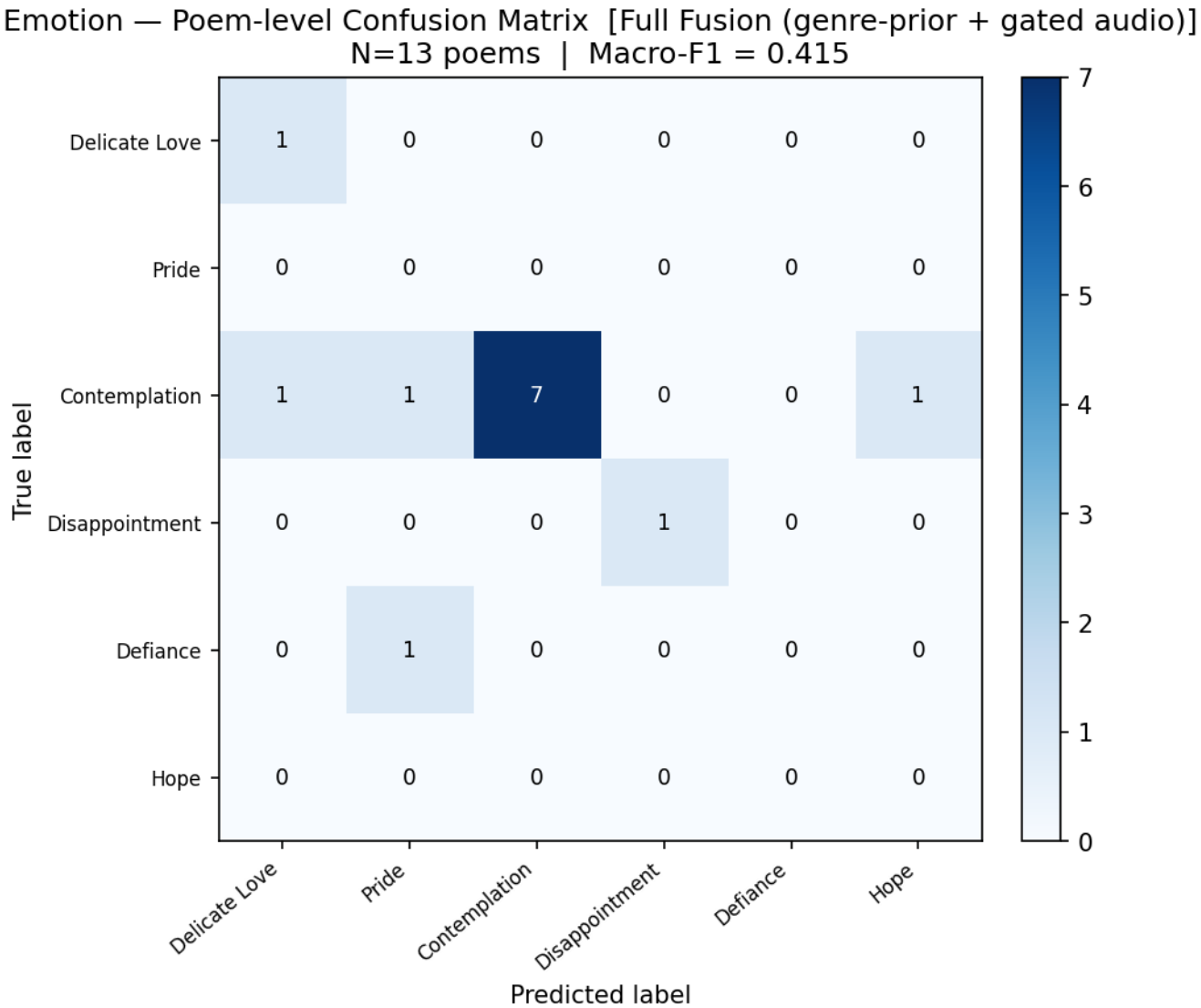

Figure 7: Emotion poem-level confusion matrix (N=13 poems, full fusion, F1=0.415). Contemplation correctly identified 7/10 times; rare emotions flagged for human review.

6.3 Arousal

Arousal MLP (22K params, MFCC features, F1=0.797). MFCC_mean_1 (spectral tilt) and RMS_mean (loudness) are the top features — physically interpretable arousal signals.

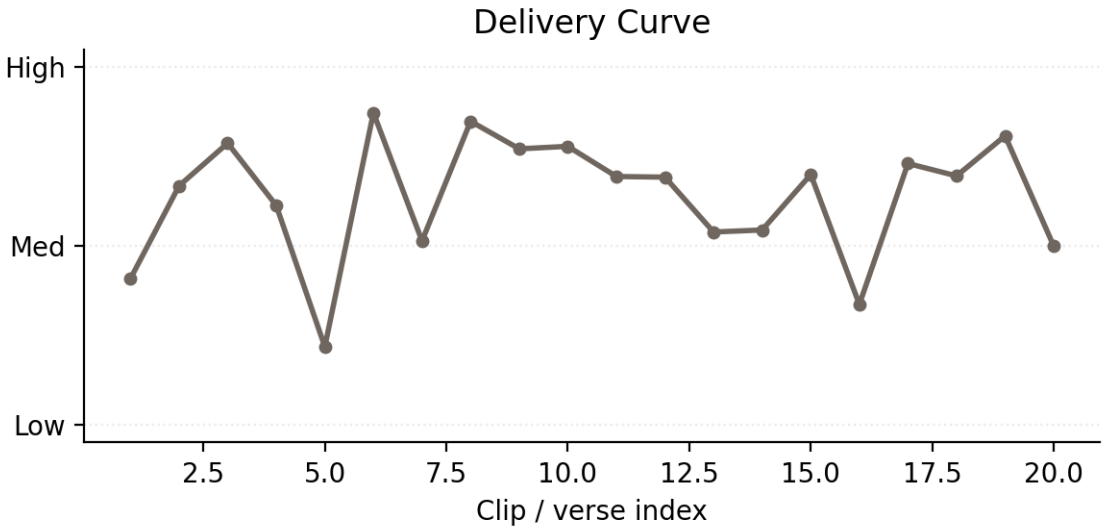

Figure 8: Arousal delivery curve — poem 0070, 20 clips. Arousal arc oscillates Medium/High with one Low dip, tracing the performer's emotional structure.

7. Key Insights

1 — The cataloguing backlog is breakable at scale, but only if genre and emotion are labelled together. A foundation cataloguing 50 Nabati recordings per week spends ~AED 12,000/week on specialist labour. Nabat-AI's emotion partial-credit of 0.862 means 86 of every 100 poems can be auto-tagged without human review — cutting weekly cost to ~AED 1,700, an 86% reduction, while routing only the genuinely ambiguous 14% to experts. This accuracy depends entirely on joint genre+emotion training: a genre-only system achieves poem F1=0.132 on the same data; adding the emotion head at zero inference cost raises genre clip F1 to 0.543 (4.1× improvement). The implication for any future labelling project: annotating both attributes from day one costs less than annotating either attribute twice.

2 — Investing in Gulf-dialect speech data is a strategic priority, not a model problem. Seven fine-tuning strategies — data curricula, architecture freezes, augmentation, optimisers, domain replay — all degraded transcription below the zero-shot baseline. This is not a technology failure; it is a data-scale signal. Safe dialect fine-tuning requires ~30K labelled utterances — roughly a 10-year collection horizon at current community-recording rates. The actionable decision is: use zero-shot transcription now (73% word accuracy is operationally sufficient), and direct resources toward structured recording programmes and community partnerships. Spending on fine-tuning experiments will not yield returns until the corpus is an order of magnitude larger.

3 — A text-only emotion system would silently mislabel the majority of this archive. In Nabati oral tradition, a grief poem performed with vigorous vocal energy is deliberate artistry. Across 3,340 clips, text and audio emotion disagree in 60.6% of cases — a near-zero inter-modality agreement (κ=0.098). Any standard NLP pipeline applied to this corpus would produce labels that conflict with what a listener actually hears in six out of ten poems, without any visible error signal. The gated fusion system cuts this mislabelling rate to 30.8% by deploying audio only to break close calls. For an archive patron searching for poems of grief, the difference is between finding recordings that genuinely sound like grief and finding recordings that merely use grief vocabulary — the distinction between a useful cultural resource and a misleading one.

8. Limitations & Reproducibility

Limitations:

- Test set: 13 poems / 7 poets — CIs are wide; no poem-level significance claims.
- Hikma = 47% of test — macro-F1 is structurally suppressed.
- Corpus is 36 poets, predominantly male; may not generalise to female Nabati poets or other GCC regions.
- DMS rate 30.8% is irreducible without subjective "intended emotion" ground-truth.

Reproducibility:

Seeds: `torch.manual_seed(42)`, `np.random.seed(42)`, `random.seed(42)`. All `from_pretrained` calls use `local_files_only=True`. Hardware: Apple M-series (MPS auto-selected); CPU fallback on any platform. `just train` ~4–6 h; `just evaluate` ~5 min; 52 tests pass.

References

Radford et al. (2022). *Robust speech recognition via large-scale weak supervision.* arXiv:2212.04356. — Whisper. | Antoun et al. (2020). *AraBERT.* arXiv:2003.00104. | Farahani (2021). *AraPoem-BERT.* HuggingFace:

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